

## Worksheet

For each problem construct an integral which solves the given problem.

### Volume

1. Find the volume of the solid with parabolic base given by the region bounded between the  $y$ -axis and the curve  $y^2 = 4 - x$  whose cross-sections (perpendicular to the  $x$ -axis) are squares.
2. Let  $R$  be the bounded region in the first quadrant bounded between the curves  $y = \frac{4}{x+1}$ ,  $y = 1$ ,  $y$ -axis.
  - A. Find the volume of the solid of revolution generated by revolving  $R$  about the  $x$ -axis
  - B. Find the volume of the solid of revolution generated by revolving  $R$  about the line  $y = -2$
  - C. Find the volume of the solid of revolution generated by revolving  $R$  about the line  $y = 6$
  - D. Find the volume of the solid of revolution generated by revolving  $R$  about the  $y$ -axis
  - E. Find the volume of the solid of revolution generated by revolving  $R$  about the line  $x = -3$
  - F. Find the volume of the solid of revolution generated by revolving  $R$  about the line  $x = 8$
3. Let  $R$  be the bounded region bounded between the curves  $y = 3x - \frac{x^2}{2}$ ,  $y = x$ .
  - A. Find the volume of the solid of revolution generated by revolving  $R$  about the  $x$ -axis
  - B. Find the volume of the solid of revolution generated by revolving  $R$  about the line  $y = -2$
  - C. Find the volume of the solid of revolution generated by revolving  $R$  about the line  $y = 6$
  - D. Find the volume of the solid of revolution generated by revolving  $R$  about the  $y$ -axis
  - E. Find the volume of the solid of revolution generated by revolving  $R$  about the line  $x = -3$
  - F. Find the volume of the solid of revolution generated by revolving  $R$  about the line  $x = 8$